

Ayers AskDocs Phase 5: Composite vs Quality-only vs Empathy-only

Human healthcare-professional labels only. Behavior policy: physician response. Target policy: logged ChatGPT response. n=195, bootstrap=100, seed=0.

Composite = mean(normalized quality, normalized empathy). Quality-only and empathy-only use one normalized dimension as Y.

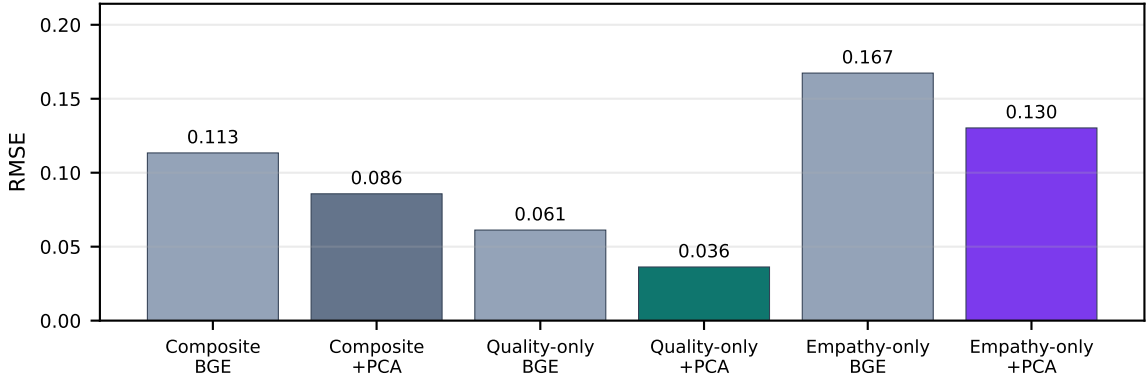
Full estimator table

Reward	Config	Est.	V_b	V_agent	Effect	V_hat	Bias	RMSE	95% CI	Cov
Composite	BGE-M3	DM	0.4254	0.7233	+0.2979	0.5537	-0.1696	0.1832	[0.502, 0.593]	no
Composite	BGE-M3	MIPS	0.4254	0.7233	+0.2979	0.6193	-0.1040	0.1134	[0.555, 0.696]	no
Composite	BGE-M3	OffCEM	0.4254	0.7233	+0.2979	0.5551	-0.1682	0.1826	[0.503, 0.593]	no
Composite	BGE-M3 + concat-PCA	DM	0.4254	0.7233	+0.2979	0.5593	-0.1640	0.1611	[0.531, 0.597]	no
Composite	BGE-M3 + concat-PCA	MIPS	0.4254	0.7233	+0.2979	0.6464	-0.0769	0.0857	[0.588, 0.716]	no
Composite	BGE-M3 + concat-PCA	OffCEM	0.4254	0.7233	+0.2979	0.5607	-0.1626	0.1603	[0.532, 0.597]	no
Quality-only	BGE-M3	DM	0.5641	0.7829	+0.2188	0.6914	-0.0915	0.1102	[0.635, 0.720]	no
Quality-only	BGE-M3	MIPS	0.5641	0.7829	+0.2188	0.7304	-0.0526	0.0612	[0.679, 0.776]	no
Quality-only	BGE-M3	OffCEM	0.5641	0.7829	+0.2188	0.6926	-0.0903	0.1098	[0.635, 0.720]	no
Quality-only	BGE-M3 + concat-PCA	DM	0.5641	0.7829	+0.2188	0.6983	-0.0846	0.0857	[0.668, 0.727]	no
Quality-only	BGE-M3 + concat-PCA	MIPS	0.5641	0.7829	+0.2188	0.7587	-0.0242	0.0363	[0.706, 0.801]	yes
Quality-only	BGE-M3 + concat-PCA	OffCEM	0.5641	0.7829	+0.2188	0.6998	-0.0831	0.0848	[0.669, 0.727]	no
Empathy-only	BGE-M3	DM	0.2868	0.6637	+0.3769	0.3928	-0.2709	0.2655	[0.358, 0.459]	no
Empathy-only	BGE-M3	MIPS	0.2868	0.6637	+0.3769	0.5082	-0.1555	0.1674	[0.420, 0.619]	no
Empathy-only	BGE-M3	OffCEM	0.2868	0.6637	+0.3769	0.3964	-0.2673	0.2647	[0.359, 0.460]	no
Empathy-only	BGE-M3 + concat-PCA	DM	0.2868	0.6637	+0.3769	0.4423	-0.2214	0.2159	[0.402, 0.502]	no
Empathy-only	BGE-M3 + concat-PCA	MIPS	0.2868	0.6637	+0.3769	0.5423	-0.1214	0.1303	[0.466, 0.652]	no
Empathy-only	BGE-M3 + concat-PCA	OffCEM	0.2868	0.6637	+0.3769	0.4429	-0.2208	0.2153	[0.404, 0.502]	no

Ground truth by reward

Reward	V_b	V_agent	Effect
Composite	0.4254	0.7233	+0.2979
Quality-only	0.5641	0.7829	+0.2188
Empathy-only	0.2868	0.6637	+0.3769

MIPS RMSE comparison



Takeaways

- MIPS is best for all three reward definitions.
- Learned concat-PCA improves MIPS RMSE every time.
- Quality-only is easiest: MIPS RMSE 0.0612 -> 0.0363.
- Empathy-only is hardest: MIPS RMSE 0.1674 -> 0.1303.
- Only quality-only + learned MIPS covers V_agent.